Curriculum Vitae – Le Hoang Vu – AI/ML Engineer

Personal Information

Le Hoang Vu
Ward 9, Ho Chi Minh City, Viet Nam
llehoangvum10@gmail.com
github.com/PrORain-HCMUS
linkedin.com/in/lehoangvu
kaggle.com/lehoangvu

Professional Objective

AI/ML engineer and researcher with a strong mathematical foundation and a genuine enthusiasm for research in Artificial Intelligence and Machine Learning. My recent interests and research directions center on *Generative AI (GenAI)*, *Natural Language Processing (NLP)*, and *Reinforcement Learning (RL)* – building intelligent systems that learn from data, reason over language, and adapt to complex real-world environments. I aim to bridge rigorous research with practical, production-grade solutions that deliver measurable impact.

Experience

AI Engineer *Nov 2025 - Present (1 year)*
Bosch Global Software Technologies Vietnam
Ho Chi Minh City, Vietnam - On-site

Mobility Manufacturing Agent (MMA)

Description : Enterprise GenAI assistant helping Bosch plant operators troubleshoot, document, and reuse solutions across plants.
Role : Backend / AI Engineer | **Technologies :** FastAPI, ChatXDC, LangGraph, LangChain, Azure OpenAI, Qdrant, Azure Cosmos DB, Docker, Azure App Service

- Built a *RAG-based troubleshooting assistant* with expert-assist mode over past issues and technical docs – cutting troubleshooting time ∼**35%**.
- Built a guided *documentation assistant* turning conversations into structured issue reports filed to shift-log and ticketing systems – reducing manual effort ∼**80%**.
- Designed a multi-plant setup with per-site data isolation and voice-to-text capture.

Result : Live across **4** Bosch plants at >**90%** user adoption – winner of the *RB Digital Award 2025 for Efficiency*, **3rd place (Process Management)** at the *Handelsblatt Best of Technology Award 2026*, and a flagship initiative in the Bosch Mobility AI Hub.

Competence Development Roadmap Chatbot (CDR Bot)

Description : Enterprise AI chatbot recommending personalized learning paths and courses for employees.
Role : Backend Developer | **Technologies :** FastAPI, LangChain, LangGraph, MongoDB, Redis, Azure OpenAI, Azure Cosmos DB, Weviate

- Designed a *multi-agent system* (LangGraph) for learning-path planning, course search, and analytics – serving **1,000**+ employees.
- Built a secure *FastAPI* backend (Azure AD, rate limiting) with *vector search* retrieval at ∼**90%** accuracy.
- Shipped an admin dashboard tracking usage and engagement.

Result : Cut course-discovery time ∼**60%** and lifted course enrollment ∼**25%**.

PM Assistant

Description : Multi-agent AI system for project management across Jira and Azure DevOps.
Role : AI Engineer | **Technologies :** LangGraph, FastAPI, ChromaDB, Azure OpenAI, MLflow

- Designed a *multi-agent system* (LangGraph) with **5** agents for reading, writing, analysis, search, and chat across the three platforms.
- Built a *hybrid search engine* combining database queries, vector search, and keyword ranking over **100k**+ work items at ∼**92%** success.
- Added automated summarization and performance monitoring across the agents.

Result : Shifted teams from reactive tracking to proactive insights, cutting issue-triage time ∼**40%**.

Technical Skills

- **AI/Machine Learning :** TensorFlow, PyTorch, Computer Vision, LangGraph, LangChain.
- **Programming Language :** Python, C++, Java, TypeScript/JavaScript.
- **Backend & API :** FastAPI, RESTful API Design, OOP, ChromaDB, Weviate, Qdrant
- **Data & Infrastructure :** Docker, Git, Google Cloud Platform, Azure Cosmos DB.
- **Other Tools & Platforms :** LaTeX, Typst, Streamlit, Gradio, VSCode, IntelliJ IDEA.

Projects

Vision Walk github.com/PrORain-HCMUS/VisionWalk
Description : A mobile AI assistant aiding visually impaired users through real-time voice guidance.
Role : Back-end Developer | **Team :** 4 members | **Technologies :** Gemini 1.5, TensorFlow Lite, PyTorch Mobile, Google Cloud STT/TTS, SQLite, React Native, FastAPI

- Applied *Gemini 1.5* for real-time object and traffic-sign detection at <**1s** latency.
- Built bidirectional voice interaction (*Google Cloud STT/TTS*) for navigation and audio warnings.
- Developed the *React Native* app and *FastAPI* backend with GPS routing and SQLite history.

Result : AI mobility assistant for the visually impaired – scored a perfect **10/10** in the Introduction to NLP course, showcased at the HCMUS x VNG Job Fair 2025, and under review by Ho Chi Minh City for large-scale deployment.

VuDa-Chatbot-o2 github.com/PrORain-HCMUS/VuDa-Chatbot-o2
Description : A lightweight Gradio RAG chatbot unifying Deep Research, Data Analysis, Mindmap visualization, and OCR.
Role : Full-stack | **Team :** 1 member | **Technologies :** Gradio, Docling, LangChain, Pandas, Matplotlib, Google Search API, RAG, OCR

- Built a RAG chatbot with **4** retrieval modes (hybrid, graph, naive, agent) over PDF/CSV uploads.
- Added a *Deep Research* mode (Google Search + evidence scraping) and in-chat *Data Analysis* (pandas/matplotlib EDA).
- Created *Mindmap* generation and an *Evidence Inspector* (pin/exclude) for iterative refinement.

Result : Unified AI assistant combining RAG, web research, data analysis, and visualization – cutting document research time ∼**50%**.

Publications

Hoang-Vu Le, Tuong-Bach-Hy Nguyen, Bac Le *2026*
"*CONGA : Continual Neural Gated Architecture for Long-History Sequential Recommendation*"
ACM RecSys 2026 – 20th ACM Conference on Recommender Systems, Main Track (Minneapolis, Minnesota, USA) | First & corresponding author | *Accepted (Oral)*
Diagnoses two structural failure modes in sequential recommenders – positional degradation and hard context limits on long histories – and resolves them via length-robust encoding and neural long-term memory : state-of-the-art on **14 of 16** accuracy metrics across four benchmarks, with up to +**18.7%** **HR@5** on the longest-history dataset, powered by custom CUDA kernels for a **10x** faster training pipeline.

Hoang-Vu Le, Chau-Minh-Tri Vu, Truong-Giang Tran, Tuong-Bach-Hy Nguyen, Phuc-Lu Le *2026*
"*ClaUSurf@COLIEE 2026 Task 2 : A Two-Stage Retrieve-then-Reason Approach for Legal Case Entailment*"
COLIEE 2026 – Competition on Legal Information Extraction and Entailment (Singapore) | Co-First & corresponding author | *Accepted*
Perform a two-stage retrieve-then-reason pipeline where a fine-tuned Qwen-Embedding-3 retriever (**Recall@5** = **93%**) feeds an LLM entailment classifier trained via off-policy knowledge distillation (best run **F1** = **0.388**).

Hoang-Vu Le, Tuong-Bach-Hy Nguyen, Huu-Sang Le, Tien-Dat Truong, Bac Le *2026*
"*Not All Residuals Are Equal : Doubly-Stochastic Mixing Guarantees Balanced Gradient Flow in Multi-Stream Transformers*"
RET 2026 | First & corresponding author
Proves doubly-stochastic mixing is the necessary and sufficient condition for balanced gradient flow across parallel residual streams ; grounds the result in Riemannian geometry and validates on four sequential-recommendation benchmarks.

Certifications

- **International English Language Testing System (IELTS)** – *Band 7.5* — *2022*
- **Iranian Geometry Olympiad** *Certification of Jury* — *2024*
- **Modern Natural Language Processing in Python (Udemy)** — *2024*
  Certificate of Completion Certificate
- **Introduction to Android Mobile Application Development (Coursera)** — *2024*
  Certificate of Completion Certificate
- **Machine Learning Specialization (Coursera)** — *2024*
  Certificate of Completion Certificate

Achievements and Awards

- **Third Final Mathematical Cup (FMC)** - *Bronze Medal* — *2021*
- **Traditional 30/4 Olympic – Grade 11 Maths** – *Silver Medal* — *2021*
- **Odon Vallet Scholarship** — *2021*
- **Lam Dong Provincial Mathematics Competition** – *Second Prize* — *2021, 2022*
- **Iranian Geometry Olympiad - Grade Advanced** – *Aluminium Compass* — *2022*
- **Act In Space International Hackathon 2026** – *International Finalist* — *2026*
  Ranked 1st Place at Ho Chi Minh City regional round and National Final
- **Euréka Student Scientific Research Award (28th)** - *Nominated (In progress)* — *2026*

Education

University of Science – Ho Chi Minh City — *09/2022 – present*
Major : Artificial Intelligence
Attended in Ho Chi Minh City's AI Challenge 2024, Zalo AI Challenge 2025, Act In Space International Hackathon 2025

Bao Loc High School for the Gifted — *07/2019 – 07/2022*
Major : Mathematics
Achieved excellent student title and awarded the highest scholarship for all 3 years. Qualified for the province's Mathematics Olympiad selected team (2021, 2022)